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PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY

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Problem Statement & Objectives

Problem Statement

Existing UAV planners pick a route from the Pareto front with a payload-blind rule (entropy, crowding distance) or a weighted sum fixed before the search, so every payload change needs expert re-tuning and a fresh search. Given a stored front and payload tags (fragility F, weight class W, urgency U, priority P), select the best route with no re-tuning and no re-search, in under 1 s.

Objectives

1. Precompute an exact Pareto front (NAMOA*) over a 3-D airspace graph
2. Model terrain, buildings, no-fly zones and direction-aware wind per edge
3. Map payload tags to selection weights with an explicit formula
4. Benchmark Tier 1 against A* and MOPSO (HV, IGD, spread)
5. Show Tier 2 picks suit the payload, incl. non-extreme routes, in <1 s
6. Export the route as an ArduPilot mission; verify in Gazebo + SITL

Related work and research gap

Author / Year	Proposed Method	Key Findings	Limitations
Chokkalingam & Ganesan (2026)	POPSO with Pareto-entropy archive	Well-spread fronts; better path efficiency and safety	Final pick by hand-set weighted sum; wind is speed-only; no payload
Ren et al. (2019)	CDNSGA-II over an octree city model	Spread-out, collision-free distance/safety paths	Pick by crowding distance only; no mission context or wind
Deng et al. (2026)	MS-INSGA2: adaptive operators + local search	Beat 11 algorithms on 3-D planning tasks	No payload context; wind left to future work; front rebuilt per task
Mandow & Pérez de la Cruz (2010)	NAMOA*: label-based multi-objective A*	Exact Pareto front with admissible heuristics	Says nothing about which route to fly
Tsoutsouras et al. (2020)	Mass- and speed-aware power model, multi-drop planning	Near-optimal on-board trajectories in ~320 ms	One trajectory, no front; mass is physical, not a selector

Research gap

Gap 1 No payload-aware selection: fixed rule, hand-tuned sum, or none
Gap 2 No front reuse: search is re-run whenever the mission changes
Gap 3 No explicit mapping from payload description to trade-off
Gap 4 Wind direction not modelled per edge; fronts carry no weather record

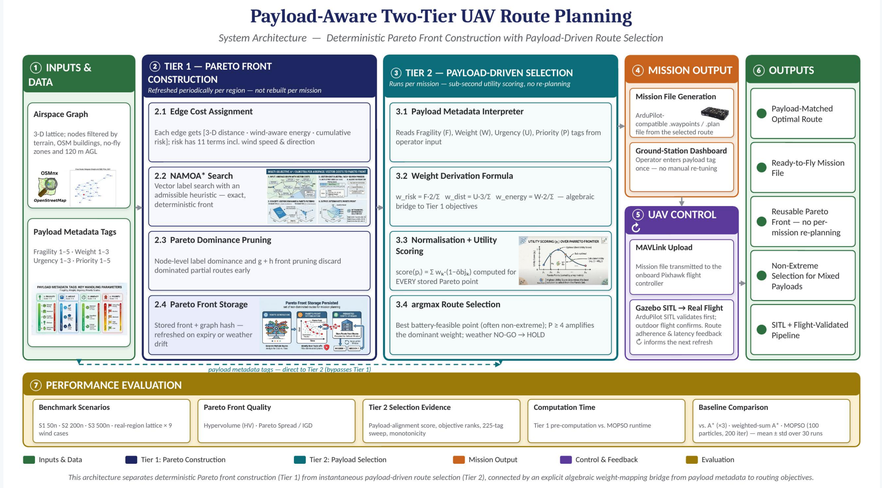
Our work: Tier 1 builds an exact front once per region and weather window; Tier 2 turns payload tags into weights and picks the best route from that stored front.

Evaluation Metrics

Reference front = non-dominated union of every route found by every method; objectives min–max normalised.

Metric	What it measures	Better
Hypervolume (HV)	Objective space the front dominates, up to reference point [1.1, 1.1, 1.1]	Higher
IGD	Mean distance from each reference route to the nearest found route	Lower; 0 = exact
Spread Δ	How evenly the routes cover the front out to its extremes	Lower
Alignment A	Σ w_k (1 − (rank_k − 1)/(K − 1)): rank-based fit of a pick to its weights	Higher
Latency / monotonicity	Tier 2 time per pick; raising F, U or W never raises risk, distance or energy (225 tags)	< 1 s; 0 violations
Cross-track RMS / landing offset	Flown SITL track vs planned route; touchdown distance from the pad	Lower (m)

Methodology



Tier 1 (once per region and weather window): every edge gets a cost vector [distance, wind-aware energy, 11-component risk]; NAMOA* finds all non-dominated routes and stores the front with its weather snapshot and expiry.

Tier 2 (per mission, <1 s): w_risk = 2F/S, w_dist = 3U/S, w_energy = 2W/S, S = 2F + 3U + 2W (P ≥ 4 boosts the largest weight by 10%); score(p) = Σ w_k (1 − normalised objective_k); route = argmax over the battery-feasible front, else HOLD.

Output & evaluation: the route is exported as a QGC WPL 110 / .plan mission, uploaded over MAVLink and flown in ArduPilot SITL + Gazebo; Module 5 scores every run with the metrics on the left.

Functional Modules and Dataset Description

Modules

M1 Airspace Graph Builder: 3-D lattice; wind-aware energy + 11-component risk
M2 Tier 1 Pareto Engine: NAMOA* + front store with drift-based refresh
M3 Baselines: A* (distance / energy / risk), weighted-sum A*, MOPSO
M4 Tier 2 Payload Interpreter: tags → weights → argmax; battery & weather HOLD
M5 Evaluation: HV, IGD, spread, alignment score, 225-tag sweep
M6 Mission Export & Dashboard: QGC WPL 110 / .plan files, Tkinter UI
M7 Simulation: Gazebo Harmonic + ArduPilot SITL, flown-track metrics

Dataset Description

Synthetic graphs (seeded): S1 50, S2 ~200, S3 500 nodes; wind 12–28 km/h
Real-region lattice: 6 km box, 200 m spacing → 893 nodes, 6,752 directed edges
Sources: Open-Meteo weather, Copernicus DEM, OpenStreetMap (Overpass)
Payload tags: 5 × 3 × 3 × 5 = 225 combinations, 4 reference presets
Simulation bundle: 1.5 km frozen region (terrain, buildings, roads, pads)

Results and Analysis

Tier 1 vs baselines (first pass; HV higher, IGD lower is better)

Scenario	Method	Routes	Time (s)	HV	IGD
S2 (200)	**NAMOA* (Tier 1)**	**16**	0.162	**1.2065**	**0.000**
S2 (200)	MOPSO	12	1.505	1.2061	0.017
S2 (200)	A* energy	1	<0.001	1.083	0.648
S3 (500)	**NAMOA* (Tier 1)**	**115**	33.761	**1.1235**	**0.000**
S3 (500)	MOPSO	38	4.326	1.1146	0.043
S3 (500)	A* energy	1	0.002	0.674	0.442

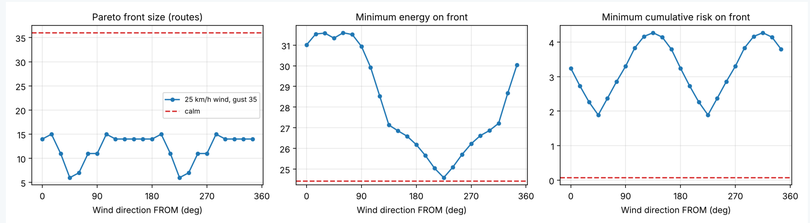
NAMOA* recovers the whole reference front (IGD 0) with the best HV; MOPSO finds only 12 and 38 routes and A* one. Exactness costs 33.8 s on S3, paid once per region and weather window.

Tier 2 picks on the S3 front (115 routes)

Preset (F, W, U, P)	Dist	Energy	Risk	Risk rank	Point type	Latency
Fragile medical (5,1,1,3)	41.02	45.86	**1.92**	25	non-extreme	0.144 ms
Emergency medicine (1,1,3,5)	**37.11**	44.71	6.56	94	non-extreme	0.140 ms
Disaster relief (1,3,1,2)	37.52	**44.27**	5.63	88	min-energy	0.136 ms
Mixed cargo (3,2,2,3)	39.90	45.06	2.74	32	non-extreme	0.136 ms

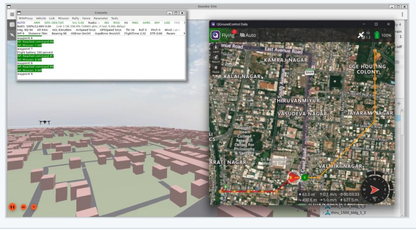
Four presets get four different routes, three non-extreme. Fragile medical flies 10.5% further than emergency medicine for 71% less risk; disaster relief takes the min-energy route.

Wind direction sweep on S2 (25 km/h, gust 35; dashed = calm air)



Panels: front size, minimum energy and minimum cumulative risk as the wind turns 360°. Direction alone moves the cheapest route's energy by 27.5%; opposite winds give equal risk (same crosswind) but different energy (headwind vs tailwind), and the front shrinks from 36 routes in calm air to 6–15.

Tier 2 mission flown in Gazebo + ArduPilot SITL



Gazebo city world from OSM buildings (left), MAVProxy console reaching each waypoint and the QGroundControl mission view (right).

Upload, read-back, AUTO, landing and disarm all succeeded: **cross-track RMS 0.11–0.13 m, landing offset 0.01–0.02 m.**

Conclusion and future work

Conclusion: Tier 1 gives an exact front (IGD 0) once per region and weather window; Tier 2 turns payload tags into distinct, payload-appropriate routes in ~0.14 ms, and those routes fly in SITL within 0.13 m of the plan. Wind direction must be modelled per edge.

Contribution: an explicit formula from payload tags to Pareto selection weights over a reusable exact front: no expert re-tuning and no re-search per mission.

Future work: full MOPSO benchmark (100 × 200 × 30 runs, significance tests), wind in Gazebo and battery calibration; Phase II on a Pixhawk F450 with Raspberry Pi 5 running Tier 2 on board.